A disproof of the Twin Primes conjecture proving also that there are no Siegel Zeros.

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Abstract :

Motivated by the previous successful disproof of the Collatz conjecture, I present this new short article that follows a new approach in order to disprove the "Twin Primes conjecture". Disproving this conjecture is a great event for number theory and prime distribution since such a disproof implies also that there are no Siegel zeros and this is a result that interests even famous mathematicians. Hence, this article will await the critics and judgments of the lovers of the number theory before any acceptance by the scientific community.

Keywords : primes, Twin-Primes, upper-bound, infinity, conjecture, disproof.

1. Introduction :

After a successful disproof of the Collatz conjecture [1], efforts were made in order to treat other famous conjectures such as the "Twin Primes conjecture". However, the previous work in this purpose [2] didn't give enough results and was useful only for reconsidering the asymptotic numbers related to Sieve theory which may be promising for other researches [3].
The "Twin Primes conjecture" obliges to prove if there are infinitely many twin primes.
The correctness of this conjecture implies also that there are definitely pairs of twin-primes approaching infinity [4, 5].

The present short article could successfully disprove the "Twin Primes conjecture" by following a new approach regarding the distance between twin primes pairs and by using the upper-bound of the number of twin primes up to the number ***y*** and that is conventionally expressed as $\pi_2(y)$. We should remind that this upper-bound was demonstrated by the professor Viggo Brun in one of his famous works [4, 5].
As a direct consequence of this disproof, we conclude that there are no Siegel zeros thanks to the theorem of Heath-Brown [6]. This result has therefore an extraordinary impact on analytic number theory, prime distribution theory and on quadratic Dirichlet L-functions.

Let's start our approach by demonstrating a useful equation regarding the sum of pairs of twin primes.
Let's consider that ***X*** is a strictly positive natural number with ***X>5***. (1)
Let's consider that ***p*** is a prime number with ***p>5.*** (2)
Let's consider that ***p*** and ***p-2*** are twin primes. (3)

We consider also that: $X=(p-2)+Y$. (4)

And thus, we have this first: $X\times p=(p-2)\times p+Y\times p$. (5)

Let's define the strictly positive natural numbers j_i and k_i such that $j_{i+1}=j_i+k_i$ where ***i*** is any strictly positive natural number and such that: $(p-2)\times p=(6\times j_1-1)\times(6\times j_1+1)=36\times j_1{}^2-1$ (6)

Furthermore, we know that all the primes bigger than ***3*** are of the form $6\times j+1$ or $6\times j-1$ where ***j*** is a strictly positive natural number. (7)

Let's consider also that the numbers: $p+6\times\sum_{i=1}^{N}k_i$ and $p-2+6\times\sum_{i=1}^{N}k_i$ are always a pair of twin primes for any strictly positive natural number ***N***.

We can get this second equation:

$$X\times(p+6\times k_1)=((p+6\times k_1)-2)\times(p+6\times k_1)+Y\times(p+6\times k_1)-6\times k_1(p+6\times k_1) \quad (8)$$

We consider in the following proofs that the numbers k_i have exact values but can approach infinity.

We remark that:

$$((p+6\times k_1)-2)\times(p+6\times k_1)=(6\times j_1+6\times k_1-1)\times(6\times j_1+6\times k_1+1)=(6\times j_2-1)\times(6\times j_2+1)=36\times j_2{}^2-1 \quad (9)$$

And we have also this third equation:

$$X\times(p+6\times k_1+6\times k_2)=((p+6\times k_1+6\times k_2)-2)\times(p+6\times k_1+6\times k_2)+Y\times(p+6\times k_1+6\times k_2)$$
$$-(6\times k_1+6\times k_2)(p+6\times k_1+6\times k_2) \quad (10)$$

We remark similarly that: $((p+6\times k_1+6\times k_2)-2)\times(p+6\times k_1+6\times k_2)=36\times j_3{}^2-1$ (11)

If we make a sum of ***N*** equations calculated similarly by adding the numbers $6\times k_i$, we conclude the following equation:

$$X\times(N\times p+6\times\sum_{i=1}^{N-1}(N-i)\times k_i)=36\times\sum_{i=1}^{N}j_i^2-N+Y\times(N\times p+6\times\sum_{i=1}^{N-1}(N-i)\times k_i)$$
$$-6\times p\times\sum_{i=1}^{N-1}(N-i)\times k_i-36\times\sum_{i=1}^{N-1}(\sum_{L=1}^{i}k_L)^2 \quad (12)$$

We conclude that: $$X=\frac{36\times(\sum_{i=1}^{N}j_i^2-\sum_{i=1}^{N-1}(\sum_{L=1}^{i}k_L)^2)-N+N\times p^2}{N\times p+6\times\sum_{i=1}^{N-1}(N-i)\times k_i}-p+Y \quad (13)$$

We remark that: $$\frac{36\times(\sum_{i=1}^{N}j_i^2-\sum_{i=1}^{N-1}(\sum_{L=1}^{i}k_L)^2)-N+N\times p^2}{N\times p+6\times\sum_{i=1}^{N-1}(N-i)\times k_i}-p=p-2 \quad (14)$$

And thus, we have:

$$36\times\left(\sum_{i=1}^{N} j_i^{\,2}-\sum_{i=1}^{N-1}\left(\sum_{L=1}^{i} k_L\right)^2\right)=2\times(p-1)\times\left(N\times p+6\times\sum_{i=1}^{N-1}(N-i)\times k_i\right)+N-N\times p^2 \quad (15)$$

Since $(p-1)^2=36\times j_1^{\,2}$ and after simplification, we get: (16)

$$36\times\left(\sum_{i=1}^{N} j_i^{\,2}-\sum_{i=1}^{N-1}\left(\sum_{L=1}^{i} k_L\right)^2\right)=36\times j_1^{\,2}\times N+72\times j_1\times\sum_{i=1}^{N-1}(N-i)\times k_i \quad (17)$$

And thus, we get the following important equation:

$$\sum_{i=2}^{N} j_i^{\,2}=(N-1)\times j_1^{\,2}+2\times j_1\times\sum_{i=1}^{N-1}(N-i)\times k_i+\sum_{i=1}^{N-1}\left(\sum_{L=1}^{i} k_L\right)^2 \quad (18)$$

We have also: $j_{i+1}-j_i=k_i$ consequently: $\sum_{L=1}^{i} k_L=j_{i+1}-j_1$. (19)

And thus: $\sum_{i=1}^{N-1}\left(\sum_{L=1}^{i} k_L\right)^2=\sum_{i=1}^{N-1}(j_{i+1}-j_1)^2=\sum_{i=2}^{N}(j_i-j_1)^2$ (20)

Consequently:

$$\sum_{i=2}^{N} j_i^{\,2}-\sum_{i=1}^{N-1}\left(\sum_{L=1}^{i} k_L\right)^2=\sum_{i=2}^{N}\left(j_i^{\,2}-(j_i-j_1)^2\right)=\sum_{i=2}^{N}\left(2\times j_1\times j_i-j_1^{\,2}\right)=2\times j_1\times\sum_{i=2}^{N} j_i-\sum_{i=2}^{N} j_1^{\,2} \quad (21)$$

And thus: $\sum_{i=2}^{N} j_i^{\,2}-\sum_{i=1}^{N-1}\left(\sum_{L=1}^{i} k_L\right)^2=2\times j_1\times\sum_{i=2}^{N} j_i-(N-1)\times j_1^{\,2}$ (22)

We get consequently the following equation: $\sum_{i=2}^{N} j_i=(N-1)\times j_1+\sum_{i=1}^{N-1}(N-i)\times k_i$ (23)

We consider in all this article that the cumulative chain above is a single cumulative chain that can gather all the twin primes bigger than ***3*** and we consider also that j_1 has always a finite value. Furthermore, we may think that the twin primes should be represented by more than one cumulative chain. However, we remind that the numbers k_i have exact values but can approach infinity in the proofs of this article with j_1 has a finite value and we know that all the primes bigger than ***3*** are of the form $6\times j+1$ or $6\times j-1$ where ***j*** is a strictly positive natural number, and thus all these cumulative chains can be merged in only one larger and single cumulative chain and it is the chain used in the proofs of the article and which is: $\sum_{i=2}^{N} j_i=(N-1)\times j_1+\sum_{i=1}^{N-1}(N-i)\times k_i$.

We can prove and check the same cumulative chain differently with:

$$\sum_{i=1}^{N} j_i=j_1+(j_1+k_1)+(j_1+k_1+k_2)+.....+\left(j_1+\sum_{i=1}^{N-1} k_i\right)=N\times j_1+\sum_{i=1}^{N-1}(N-i)\times k_i \quad (24)$$

Important Remark:

We can't have three consecutive odd numbers bigger than ***3*** as primes because one of these three consecutive numbers is divisible by ***3***. This is the reason why the cumulative chain above counts exactly ***N*** that is the count of all the numbers j_i that equals also the number of the pairs of twin primes. However, in the case of the Polignac's conjecture with a difference ***n>2***, we can have three or more consecutive primes with the same fixed difference ***n***. Hence, in this case, at least one prime $p_i=q$ of these primes plays two roles corresponding to two numbers j_i and j_{i+1} and should satisfy:

$$q=6\times j_i+1=6\times j_{i+1}-1 \quad \text{hence:} \quad j_{i+1}-j_i=\frac{2}{6} \quad \text{consequently} \quad k_i=\frac{1}{3} \quad \text{with} \quad p_{i+1}=q+n \qquad (25)$$

This means that we have: $p_{i+2}=q+6\times k_i=q+2$ and this contradicts the fact that: $p_{i+1}=q+n$ (26)

We conclude that the approach of this article can't prove, for the case of the Polignac's conjecture with a difference ***n>2***, the existence of a cumulative chain similar to the cumulative chain above that counts exactly ***N*** that is the count of all the numbers j_i that equals also the number of primes of difference ***n>2***.

This problem is impossible in the case of the cumulative chain above of the twin primes conjecture where we have ***n=2*** since we have always $p_{i+2}>q+4$ with $p_{i+1}=q+2$. (27)

Furthermore, in the cumulative chain above, we proved that the number ***N*** of the pairs of twin primes can be counted exactly whereas each twin prime p_i only plays a unique role corresponding to a unique number j_i and this is a characteristic of twin primes that will be very important when using an inequality of the professor Viggo Brun [4, 5] in the proof of this article since it allows a unique strict order of the numbers j_i after fixing the indices ***i*** in the cumulative chain.

2. The approach of considering a difference H bigger than the different values of k_i and the consequences on the Twin Primes conjecture:

We remark that: $$\frac{\sum_{i=2}^{N} j_i}{N\times(N-1)}=\frac{j_1}{N}+\frac{\sum_{i=1}^{N-1}(N-i)\times k_i}{N\times(N-1)} \qquad (28)$$

And we have: $$\sum_{i=1}^{N-1}(N-i)\times k_i\le\frac{N\times(N-1)}{2}\times k_{max} \quad \text{with} \quad k_{max}=Max\{(j_i-j_{i-1})\} \quad \text{and} \quad i\le N \qquad (29)$$

We should remark that k_{max} depends on ***N*** and we get: $$\frac{\sum_{i=2}^{N} j_i}{N\times(N-1)}\le\frac{j_1}{N}+\frac{k_{max}}{2} \quad . \qquad (30)$$

Let's consider that for each value of ***N*** we have: $k_{max}=(j_z-j_{z-1})=k_{z-1}$. The natural number ***z*** exists for each value of ***N.*** (31)

Let's consider now that: $H=\sum_{i=1}^{z-1} k_i$ and that $J_2=H+j_1=j_z$ and that: $J_1=j_1$ and $J_i=j_{z-2+i}$ when ***i>1*** and with $z\geq 2$. (32)

Hence, we have: $\sum_{i=z}^{N} j_i=\sum_{i=2}^{M} J_i$ with ***M=N-z+2.*** (33)

And similarly to this proven inequality: $\frac{\sum_{i=2}^{N} j_i}{N\times(N-1)}\leq\frac{j_1}{N}+\frac{k_{max}}{2}$ we can conclude that:

$\frac{\sum_{i=2}^{M} J_i}{M\times(M-1)}\leq\frac{J_1}{M}+\frac{H}{2}$ and ***H*** exists for each value of ***M*** with $H=J_2-j_1$. (34)

This is true because we will choose ***M*** such that ***M<L*** where ***L*** is the smallest natural number implying that: $H<J_L-J_{L-1}$. (35)

Consequently: $\frac{\sum_{i=3}^{M} J_i}{M\times(M-1)}\leq\frac{J_1}{M}+\frac{J_2-j_1}{2}-\frac{J_2}{M\times(M-1)}$ (36)

Hence: $\frac{\sum_{i=3}^{M} J_i}{M\times(M-1)}\leq\frac{J_1}{M}-\frac{j_1}{2}+J_2\times(\frac{1}{2}-\frac{1}{M\times(M-1)})$ (37)

And thus: $\frac{2\times\sum_{i=3}^{M} J_i}{2\times M\times(M-1)}-J_2\times(\frac{M\times(M-1)-2}{2\times M\times(M-1)})\leq j_1\times(\frac{1}{M}-\frac{1}{2})$ (38)

Consequently: $\frac{2\times\sum_{i=3}^{M} J_i+2\times J_2}{2\times M\times(M-1)}-J_2\times(\frac{M\times(M-1)}{2\times M\times(M-1)})\leq j_1\times(\frac{1}{M}-\frac{1}{2})$ (39)

Hence: $\frac{\sum_{i=2}^{M} J_i}{M\times(M-1)}-\frac{J_2}{2}\leq j_1\times(\frac{1}{M}-\frac{1}{2})$ (40)

Consequently, it is sufficient that we have ***M>2*** in order to have: $\frac{\sum_{i=2}^{M} J_i}{M\times(M-1)}<\frac{J_2}{2}$ (41)

We conclude that: $\frac{\sum_{i=2}^{M} J_i}{J_2\times M\times(M-1)}<\frac{1}{2}$ (42)

This means that for ***M>2*** we have: $\frac{\sum_{i=2}^{M} J_i}{J_2\times M\times(M-1)}<\frac{1}{2}$ (43)

We should also add that if it is always possible to have ***M=3*** for any possible value of J_2 , then we have always: $\frac{\sum_{i=2}^{3} J_i}{J_2 \times 3 \times (3-1)} < \frac{1}{2}$ hence: $J_2 + J_3 < 3 \times J_2$ and thus: $J_3 < 2 \times J_2$. (44)

Let's consider that it is always possible to have $j_i = J_M$ with ***M=3.*** This means also that if J_3 approaches infinity than J_2 approaches infinity too since $J_3 < 2 \times J_2$. (45)

However, since ***M=3*** is always possible, then we can reduce the value of ***H*** from $H = \sum_{i=1}^{z-1} k_i$ to $H = \sum_{i=1}^{z-2} k_i$ in order to make J_2 play the previous role of J_3 , but J_2 approaches infinity. (46)

We remind that we have: $J_2 = H + J_1 = j_z$ and $(J_i - J_{i-1}) \leq H$ for any strictly positive natural number ***i*** that is bigger than ***2*** and that is smaller or equal to ***M***. (47)

We remind also that the correctness of the "Twin Primes conjecture" implies also that there are definitely pairs of twin-primes approaching infinity [4, 5].
Repeating the process above of reducing the value of ***H*** will make all the possible values of J_2 approach infinity. And thus, we will have j_i approaching infinity for any strictly positive value of ***i*** and this is impossible.

We conclude that, if it is always possible to have ***M=3*** for any possible value of J_2 then all the possible values of J_3 should be finite and this means that the "Twin Primes conjecture" can't be true by satisfying the considerations of this article.
And thus, if there exists at least one value ***G*** of the number ***z*** where we have $J_{G+1} \geq 2 \times j_G$, then in this case the "Twin Primes conjecture" may be true.

3. The considerations regarding the Twin Primes conjecture and the contradictions leading to the disproof:

We agree that if a natural number ***x*** approaches infinity then we have j_x approaches also infinity since from the index we remark that there are infinitely many numbers j_i smaller than j_x .
However, if we consider that when j_x approaches infinity we have definitely ***x*** approaches infinity, then we can conclude also that when j_{j_x} approaches infinity we have definitely j_x approaches infinity and that when $j_{j_{j_x}}$ approaches infinity we have definitely j_{j_x} approaches infinity.

And thus, this is a logical mathematical induction of indices and we can define:
$(j,x)^0 = x$ and $(j,x)^1 = j_x$ and $(j,x)^2 = j_{j_x}$ and $(j,x)^3 = j_{j_{j_x}}$ and so on… (48)
Now, let's consider that ***x*** is finite. If the twin primes conjecture is true, then we have always $(j,x)^t$ approaches infinity when ***t*** approaches infinity even if ***x*** is finite. However, by our considered logical

mathematical induction of indices that assumes that when j_x approaches infinity we have definitely $\boldsymbol{x}$ approaching infinity, we have $(j,x)^{t-1}$ approaches infinity and also $(j,x)^{t-2}$ approaches infinity and thus we have also $(j,x)^0 = x$ is approaching infinity and this a contradiction with the fact that $\boldsymbol{x}$ is considered finite.

We conclude that, by considering that the "Twin Primes conjecture" is true, there exists a finite natural number $\boldsymbol{x}$ with j_x approaching infinity.

Let's consider that consider that the "Twin Primes conjecture" is true and that we have $\boldsymbol{x}$ is finite but j_x approaches infinity.

Also, the professor Viggo Brun proved in his valuable works [4, 5] that for some constant $\boldsymbol{E}$ we have:

$$\pi_2(y) < E \times \frac{y}{(\log y)^2} \tag{49}$$

where $\pi_2(y)$ is the number of twin primes up to $\boldsymbol{y}$.

In the cumulative chain of this article, we proved that the number $\boldsymbol{N}$ of the pairs of twin primes can be counted exactly whereas each twin prime p_i only plays a unique role corresponding to a unique number j_i and this is a characteristic of twin primes that will be very important in the proof of this article because we have a unique strict order of the numbers j_i after fixing the indices $\boldsymbol{i}$ in the cumulative chain of this article and this implies that the Brun's inequality above means also that with $\boldsymbol{y}$ is a finite positive natural number, we have:

$$6 \times j_{[\frac{\pi_2(y)-s}{2}]} + 1 < y \quad \text{with } \boldsymbol{s} \text{ is the finite number of twin primes smaller than } 6 \times j_1 - 1 \tag{50}$$

where $[\frac{\pi_2(y)-s}{2}]$ is the integer part or whole part of $\frac{\pi_2(y)-s}{2}$.

And thus, by considering that $x < [\frac{\pi_2(y)-s}{2}]$ we have also: $6 \times j_x + 1 < y$. (51)

However, we considered that j_x approaches infinity even if $\boldsymbol{x}$ is finite.
Since $\boldsymbol{y}$ is finite, then this is a contradiction that proves that we can't have a finite natural number $\boldsymbol{x}$ with $x < [\frac{\pi_2(y)-s}{2}]$ and j_x is approaching infinity.

We conclude that we have always $x \geq [\frac{\pi_2(y)-s}{2}]$ for any finite value of $\boldsymbol{y}$. (52)

However, we considered that the twin primes conjecture is true and this means that $\boldsymbol{x}$ approaches infinity. And thus, this is a contradiction with the fact that $\boldsymbol{x}$ is a finite natural number.
Consequently, we can't consider that the twin primes conjecture is true by satisfying the considerations of this article.

This means that the set of the numbers: $p + 6 \times \sum_{i=1}^{N} k_i$ and $p - 2 + 6 \times \sum_{i=1}^{N} k_i$ that are always a pair of twin primes for any strictly positive natural number $\boldsymbol{N}$ is not infinite.

We know also that the pair of twin primes that can't be expressed as the numbers: $p+6\times\sum_{i=1}^{N} k_i$ and $p-2+6\times\sum_{i=1}^{N} k_i$ for any strictly positive natural number ***N*** are only elements of a finite set.

Eureka! We can conclude then that the "Twin Primes Conjecture" is not true.

Furthermore, a very important result by Roger Heath-Brown's [6] can be stated as follows:

At least one of the two following statements is true:
1- There are no Siegel zeros.
2- There are infinitely many twin primes.

We conclude thanks to this theorem that there are no Siegel zeros.

« Quod erat demonstrandum ».

4. Summary and conclusion:

We started our work by proposing an extensive procedure in order to prove that: $\sum_{i=2}^{N} j_i^2=(N-1)\times {j_1}^2+2\times j_1\times\sum_{i=1}^{N-1}(N-i)\times k_i+\sum_{i=1}^{N-1}\left(\sum_{L=1}^{i} k_L\right)^2$ where the positive natural numbers j_i and k_i are such as $j_{i+1}=j_i+k_i$ with ***i*** is any strictly positive natural number and such as $(p-2)\times p=(6\times j_1-1)\times(6\times j_1+1)=36\times {j_1}^2-1$.

We could then simplify the equation above and get: $\sum_{i=2}^{N} j_i=(N-1)\times j_1+\sum_{i=1}^{N-1}(N-i)\times k_i$.

This cumulative chain above is a single cumulative chain than can gather all the twin primes bigger than ***3.***

We used afterwards the equation above in order to prove that if it is always possible to have ***M=3*** for any possible value of J_2 , then we have always: $\frac{\sum_{i=2}^{3} J_i}{J_2\times 3\times(3-1)}<\frac{1}{2}$ hence: $J_2+J_3<3\times J_2$ and thus: $J_3<2\times J_2$.

We remind that we have: $J_2=H+J_1=j_z$ and $(J_i-J_{i-1})\leq H$ for any strictly positive natural number ***i*** that is bigger than ***2*** and that is smaller or equal to ***M***.
We concluded that the twin primes conjecture can't be true if it is always possible to have ***M=3.***
Finally, we proved the existence of a finite value ***x*** such that by assuming that the "Twin Primes conjecture" is true then we have j_x approaches infinity.

However, we used the following upper-bound of the number of twin primes up to ***y*** expressed as:

$\pi_2(y) < E \times \frac{y}{(\log y)^2}$ for some constant ***E*** where $\pi_2(y)$ is the number of twin primes up to the positive natural number ***y.***

This upper-bound could lead to a contradiction stating that the natural number ***x*** approaches infinity even if it is considered finite before in the article. And thus, we concluded that the “Twin Primes Conjecture” is not true.

We can also conclude as an annex from Viggo Brun’s works [4, 5] that: $\sum_{i=2}^{N} \frac{1}{j_i} = R$ with ***R*** finite when ***N*** approaches infinity since we have:

$\frac{1}{p_i} + \frac{1}{p_i+2} = \frac{1}{6 \times j_i - 1} + \frac{1}{6 \times j_i + 1} = \frac{12 \times j_i}{36 \times j_i^2 - 1} > \frac{12 \times j_i}{36 \times j_i^2} = \frac{1}{3 \times j_i}$ where p_i and p_i+2 are twin primes such as $p_i = 6 \times j_i - 1$. (53)

We remind also that the “Twin Primes” conjecture is the case of the even number ***n=2*** in the Polignac’s conjecture and we can also say that as a result of this article the “Hardy-Littlewood Twin prime conjecture” that is sometimes called the quantitative form of the “Twin Primes Conjecture” is also not true. Furthermore, thanks to Roger Heath-Brown's theorem (1983), we could prove that there are no Siegel zeros.

This significant result brings about transformative improvements in estimating the ranks of elliptic curves and L-functions, while also facilitating optimal error terms in the Prime Number Theorem for arithmetic progressions. It contributes to advancements in sieve theory and class number bounds for quadratic fields, marking a major step forward in analytic number theory and the theory of prime distribution. In number theory, it's common to encounter the expression $2 \times C_2 \approx 1.32032367.....$, known as the twin-prime constant. Although the Twin Primes conjecture fails at a global level, the local expectation remains positive across finite scales. The twin-prime constant continues to serve as the accurate local-global constant for prime constellations shaped as "two primes within an interval of fixed even length," with its value remaining unchanged and precise.

Should the disproof of the Twin Primes conjecture gain acceptance within the scientific community, it would bring forth numerous implications in number theory, particularly impacting the distribution of prime numbers and related asymptotic properties. This may have repercussions for major unresolved problems like the Riemann Hypothesis, which is intricately linked to prime distribution. Moreover, any credible solution to the Riemann Hypothesis—possibly aided by Artificial Intelligence and gaining community approval—could assist in addressing Goldbach's Conjecture. This is where the twin-prime constant comes into play once more. Proving the Riemann Hypothesis would imply a highly regular distribution of primes, lending strong support to Goldbach's Conjecture by demonstrating the abundance and regular spacing of primes necessary to form the required sum of primes.

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